

ST01 Absolute BA uncertainty

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Stratum	Model	Mean BA	Context SD	95% crossed CI
OrganAMNIST / A (10×5; 4 models)	DenseNet-121	0.9904	0.0018	[0.9895, 0.9913]
OrganAMNIST / A (10×5; 4 models)	EfficientNet-B0	0.9620	0.0030	[0.9606, 0.9635]
OrganAMNIST / A (10×5; 4 models)	ResNet-18	0.9948	0.0015	[0.9940, 0.9954]
OrganAMNIST / A (10×5; 4 models)	ResNet-50	0.9942	0.0015	[0.9935, 0.9949]
OrganAMNIST / A (10×3; 4 models) [common3]	DenseNet-121	0.9901	0.0019	[0.9891, 0.9911]
OrganAMNIST / A (10×3; 4 models) [common3]	EfficientNet-B0	0.9621	0.0031	[0.9605, 0.9637]
OrganAMNIST / A (10×3; 4 models) [common3]	ResNet-18	0.9948	0.0016	[0.9938, 0.9955]
OrganAMNIST / A (10×3; 4 models) [common3]	ResNet-50	0.9945	0.0014	[0.9938, 0.9952]
OrganAMNIST / B (10×5; 4 models)	DenseNet-121	0.9921	0.0015	[0.9913, 0.9928]
OrganAMNIST / B (10×5; 4 models)	EfficientNet-B0	0.9628	0.0023	[0.9616, 0.9640]
OrganAMNIST / B (10×5; 4 models)	ResNet-18	0.9960	0.0009	[0.9956, 0.9965]
OrganAMNIST / B (10×5; 4 models)	ResNet-50	0.9962	0.0013	[0.9956, 0.9968]

Mean BA and context-level sample SD (ddof=1). Mean intervals use paired crossed split–seed resampling. The 16 strata include reused subsets and candidate/input sensitivities; rows must not be summed as independent training runs.

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Stratum	Model	Mean BA	Context SD	95% crossed CI
OrganAMNIST / B (10×3; 4 models) [common3]	DenseNet-121	0.9921	0.0016	[0.9912, 0.9929]
OrganAMNIST / B (10×3; 4 models) [common3]	EfficientNet-B0	0.9628	0.0023	[0.9615, 0.9642]
OrganAMNIST / B (10×3; 4 models) [common3]	ResNet-18	0.9960	0.0008	[0.9956, 0.9964]
OrganAMNIST / B (10×3; 4 models) [common3]	ResNet-50	0.9962	0.0013	[0.9956, 0.9969]
SIPaKMeD / A (10×5; 4 models)	DenseNet-121	0.9570	0.0092	[0.9523, 0.9621]
SIPaKMeD / A (10×5; 4 models)	EfficientNet-B0	0.9462	0.0077	[0.9421, 0.9507]
SIPaKMeD / A (10×5; 4 models)	ResNet-18	0.9659	0.0064	[0.9628, 0.9690]
SIPaKMeD / A (10×5; 4 models)	ResNet-50	0.9636	0.0076	[0.9598, 0.9675]
SIPaKMeD / A (10×3; 4 models) [common3]	DenseNet-121	0.9573	0.0100	[0.9514, 0.9632]
SIPaKMeD / A (10×3; 4 models) [common3]	EfficientNet-B0	0.9464	0.0078	[0.9419, 0.9511]
SIPaKMeD / A (10×3; 4 models) [common3]	ResNet-18	0.9654	0.0066	[0.9620, 0.9688]
SIPaKMeD / A (10×3; 4 models) [common3]	ResNet-50	0.9640	0.0077	[0.9599, 0.9689]

Mean BA and context-level sample SD (ddof=1). Mean intervals use paired crossed split–seed resampling. The 16 strata include reused subsets and candidate/input sensitivities; rows must not be summed as independent training runs.

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Stratum	Model	Mean BA	Context SD	95% crossed CI
SIPaKMeD / B (10×5; 4 models)	DenseNet-121	0.9611	0.0086	[0.9563, 0.9657]
SIPaKMeD / B (10×5; 4 models)	EfficientNet-B0	0.9476	0.0074	[0.9436, 0.9517]
SIPaKMeD / B (10×5; 4 models)	ResNet-18	0.9686	0.0063	[0.9654, 0.9716]
SIPaKMeD / B (10×5; 4 models)	ResNet-50	0.9688	0.0067	[0.9652, 0.9722]
SIPaKMeD / B (10×3; 4 models) [common3]	DenseNet-121	0.9605	0.0089	[0.9553, 0.9658]
SIPaKMeD / B (10×3; 4 models) [common3]	EfficientNet-B0	0.9473	0.0077	[0.9428, 0.9517]
SIPaKMeD / B (10×3; 4 models) [common3]	ResNet-18	0.9687	0.0049	[0.9661, 0.9712]
SIPaKMeD / B (10×3; 4 models) [common3]	ResNet-50	0.9676	0.0065	[0.9635, 0.9712]
ISIC2019 / A (10×3; 4 models)	DenseNet-121	0.4619	0.0190	[0.4514, 0.4732]
ISIC2019 / A (10×3; 4 models)	EfficientNet-B0	0.4488	0.0204	[0.4377, 0.4614]
ISIC2019 / A (10×3; 4 models)	ResNet-18	0.4652	0.0260	[0.4498, 0.4828]
ISIC2019 / A (10×3; 4 models)	ResNet-50	0.5077	0.0259	[0.4908, 0.5250]

Mean BA and context-level sample SD (ddof=1). Mean intervals use paired crossed split–seed resampling. The 16 strata include reused subsets and candidate/input sensitivities; rows must not be summed as independent training runs.

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Stratum	Model	Mean BA	Context SD	95% crossed CI
ISIC2019 / A (5×3; 4 models) [CNN control]	DenseNet-121	0.4625	0.0200	[0.4465, 0.4791]
ISIC2019 / A (5×3; 4 models) [CNN control]	EfficientNet-B0	0.4549	0.0237	[0.4363, 0.4759]
ISIC2019 / A (5×3; 4 models) [CNN control]	ResNet-18	0.4637	0.0213	[0.4465, 0.4793]
ISIC2019 / A (5×3; 4 models) [CNN control]	ResNet-50	0.5022	0.0217	[0.4867, 0.5200]
ISIC2019 / A (5×3; 5 models) [shared input]	DenseNet-121	0.4625	0.0200	[0.4467, 0.4791]
ISIC2019 / A (5×3; 5 models) [shared input]	EfficientNet-B0	0.4549	0.0237	[0.4357, 0.4759]
ISIC2019 / A (5×3; 5 models) [shared input]	ResNet-18	0.4637	0.0213	[0.4465, 0.4795]
ISIC2019 / A (5×3; 5 models) [shared input]	ResNet-50	0.5022	0.0217	[0.4869, 0.5192]
ISIC2019 / A (5×3; 5 models) [shared input]	Swin-T	0.5033	0.0314	[0.4809, 0.5325]
ISIC2019 / A (5×3; 5 models) [weight-eval]	DenseNet-121	0.4625	0.0200	[0.4469, 0.4792]
ISIC2019 / A (5×3; 5 models) [weight-eval]	EfficientNet-B0	0.4549	0.0237	[0.4365, 0.4749]
ISIC2019 / A (5×3; 5 models) [weight-eval]	ResNet-18	0.4637	0.0213	[0.4464, 0.4798]

Mean BA and context-level sample SD (ddof=1). Mean intervals use paired crossed split–seed resampling. The 16 strata include reused subsets and candidate/input sensitivities; rows must not be summed as independent training runs.

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Stratum	Model	Mean BA	Context SD	95% crossed CI
ISIC2019 / A (5×3; 5 models) [weight-eval]	ResNet-50	0.5022	0.0217	[0.4872, 0.5192]
ISIC2019 / A (5×3; 5 models) [weight-eval]	Swin-T	0.5229	0.0316	[0.5011, 0.5508]
MURA / A (10×3; 4 models)	DenseNet-121	0.7873	0.0100	[0.7805, 0.7928]
MURA / A (10×3; 4 models)	EfficientNet-B0	0.7715	0.0075	[0.7670, 0.7755]
MURA / A (10×3; 4 models)	ResNet-18	0.7762	0.0150	[0.7678, 0.7835]
MURA / A (10×3; 4 models)	ResNet-50	0.7896	0.0166	[0.7790, 0.7999]
MURA / A (5×3; 4 models) [CNN control]	DenseNet-121	0.7899	0.0088	[0.7821, 0.7957]
MURA / A (5×3; 4 models) [CNN control]	EfficientNet-B0	0.7729	0.0081	[0.7654, 0.7786]
MURA / A (5×3; 4 models) [CNN control]	ResNet-18	0.7809	0.0132	[0.7720, 0.7906]
MURA / A (5×3; 4 models) [CNN control]	ResNet-50	0.7954	0.0154	[0.7816, 0.8071]
MURA / A (5×3; 5 models) [shared input]	DenseNet-121	0.7899	0.0088	[0.7811, 0.7958]
MURA / A (5×3; 5 models) [shared input]	EfficientNet-B0	0.7729	0.0081	[0.7648, 0.7785]

Mean BA and context-level sample SD (ddof=1). Mean intervals use paired crossed split–seed resampling. The 16 strata include reused subsets and candidate/input sensitivities; rows must not be summed as independent training runs.

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Stratum	Model	Mean BA	Context SD	95% crossed CI
MURA / A (5×3; 5 models) [shared input]	ResNet-18	0.7809	0.0132	[0.7717, 0.7906]
MURA / A (5×3; 5 models) [shared input]	ResNet-50	0.7954	0.0154	[0.7816, 0.8071]
MURA / A (5×3; 5 models) [shared input]	Swin-T	0.8068	0.0082	[0.8006, 0.8130]
MURA / A (5×3; 5 models) [weight-eval]	DenseNet-121	0.7899	0.0088	[0.7815, 0.7957]
MURA / A (5×3; 5 models) [weight-eval]	EfficientNet-B0	0.7729	0.0081	[0.7649, 0.7786]
MURA / A (5×3; 5 models) [weight-eval]	ResNet-18	0.7809	0.0132	[0.7719, 0.7907]
MURA / A (5×3; 5 models) [weight-eval]	ResNet-50	0.7954	0.0154	[0.7816, 0.8071]
MURA / A (5×3; 5 models) [weight-eval]	Swin-T	0.8060	0.0057	[0.8019, 0.8108]

Mean BA and context-level sample SD (ddof=1). Mean intervals use paired crossed split–seed resampling. The 16 strata include reused subsets and candidate/input sensitivities; rows must not be summed as independent training runs.